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Modelling women's team line-ups based on effectiveness and quality

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ABSTRACT

Understanding how different player rotations may impact team performance allows basketball coaches to select effective line-ups for specific tactical scenarios. The study aimed to i) assess how different line-ups or player combinations impact a team's game performance; ii) explore the variations in line-up utilization among different national women's basketball teams; and iii) examine how the offensive efficiency of each line-up evolves during the game. Data from 3,387 ball possessions in 23 international women's basketball games were collected across four major competitions over six years. Offensive and defensive ratings, along with other features, were calculated. Then, a Markov chain model distinguished overperforming and underperforming line-ups of Chinese women's basketball team, determining long-term probabilities for each rating level. The results indicated that i) the most dominant offensive line-up of the Chinese women's basketball team, is PG-G-SF-PF-C, while G-G-F-PF-PF had the highest defensive rating; and ii) US and Australian women's basketball teams favour using line-ups with three guards, while the Chinese women's basketball team heavily relies on centre players. These results offer valuable insights for coaches regarding the performance of different line-ups in FIBA Female Basketball Competitions, optimizing line-up performance and aiding talent selection and recruitment at the international level.

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Performance analysis; team sports; playing positions; quintet; international competitions

1. Introduction

Basketball is a fast-paced and dynamic game that requires teamwork and effective communication to achieve success so the analysis of its match performance has been one of the widely investigated topics in team sports (Wen, 2019). Recent research on basketball performance analysis has covered a wide range of aspects, including but not limited to, game outcome prediction (Sikka & Devarajan, 2022; Štrumbelj & Vračar, 2012), shot quality (Harmon et al., 2021), defensive actions (Franks et al., 2015; Nunes et al., 2021), or game strategy (Noivo et al., 2022; Tian et al., 2019). One of the prominent research topics in this field is the analysis of players' line-ups and their impact on performance (Ahmadalinezhad & Makrehchi, 2020). This area of study has gained considerable attention due to its potential to improve team management and decision-making processes, as a team with excellent players who fail to cooperate with their teammates may not be as effective as a team with comparatively less competitive players who have established seamless collaboration (Group, 2021).

Data-driven sports performance analysis offers holistic insights into player and team dynamics. The box score data, employed since the mid-1900s, summarizes player statistics – minutes played, field goals, rebounds, assists, steals, blocks, and turnovers – enabling game and season evaluations. However, when assessing team performance, emphasizing on-court cohesion and communication is crucial, beyond mere aggregation of individual player achievements (Onağ & Tepeci, 2014). A previous study avoided the typical use of box score data to identify key determinants of team success in wheelchair basketball (Francis et al., 2019). Also, the concept of game possession and the quantification of a team's offensive and defensive rating become valuable, enabling identification of optimal five-player line-ups excelling in offence and defence (Kubatko et al., 2007). This information provides a more comprehensive evaluation of a team's performance and can be used by coaches to make strategic decisions about which players to use in different game contexts.

In team sports such as volleyball, football, and basketball, constant player substitutions during games introduce a dynamic element, making it challenging for coaches to manage and strategically influence the evolving state of the game (Amez et al., 2021; M.-A. Gómez et al., 2016; Silva et al., 2016). A previous study revealed the significance of well-timed substitutions, especially in the context of diverse influences across the first three quarters. However, a diminished impact was observed during the fourth quarter (M.-Á. Gómez et al., 2017). Relevant findings have indicated that some coaches prefer to make frequent substitutions to keep their players fresh and maintain a high level of energy throughout the game, while others prefer to stick with a smaller rotation of players and limit substitutions to maintain consistency and chemistry on the court (Clay & Clay, 2014).

To make effective decisions about player substitutions, coaches must consider how well the new players will perform in tandem with their teammates. However, accurately predicting the performance of certain player-position combinations can

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be challenging. Previous studies have utilized several key performance indicators, such as plus-minus, adjusted plus-minus, offensive rating, defensive rating, and net rating to evaluate the efficiency of various five-player line-ups (Grassetti et al., 2021; Kolias et al., 2022; Martonosi et al., 2023). These findings indicated that making frequent player rotations can result in multiple changes to the team's line-up on the court, which further emphasizes the importance of assessing or predicting how different players perform together. By doing so, coaches will be better equipped to make informed decisions about player substitutions and rotations, leading to improved team performance on the court.

Furthermore, the comprehension and categorization of player positions and roles are dynamic, subject to the influenced of team strategy, playing style, and individual strengths (Cui et al., 2019; Trninić & Dizdar, 2000). Firstly, player positions have traditionally comprised of point guard (PG), shooting guard (SG), small forward (SF), power forward (PF), and centre (C) (FIBA, 2022). Also, previous studies define positions differently, using labels like inside players, post players, wing, guard, forward, or big (Sampaio et al., 2006; Trninić & Dizdar, 2000). Lastly, others have suggested a more flexible approach to player positions, where players are not assigned specific roles but rather play based on their skills and abilities (Garcia-Rubio et al., 2019; Kalman & Bosch, 2020; Rangel et al., 2019). Therefore, coaches should consider a range of factors when discerning player roles and characteristics in basketball, especially in the critical decision-making process of determining the optimal line-up for each game situation.

Concurrently, the study of line-up optimization has attracted significant attention, as it could help teams strengthen their strategic of player combinations. Prior investigations have highlighted mathematical optimization techniques as effective tools for achieving line-up optimization. Francis et al. (2019) employed binary logistic regression to examine the performance of line-ups in wheelchair basketball (Francis et al., 2019). Kolias et al. (2022) utilized basketball game data and statistical modelling to analyse rotation line-up efficiency, providing insights into optimal offensive and defensive line-up for coaches in the Greek Basket League (Kolias et al., 2022). Mahrudinda et al. (2020) employed binary integer programming to optimize the line-up formation for Liverpool FC in the 2020/2021 English Premier League season, revealing the 4-3-3 formation's superior player rating effectiveness compared to the 4-2-3-1 formation, accentuating its attacking prowess and speed (Mahrudinda et al., 2020). In conclusion, these studies underscore the pivotal role of mathematical optimization techniques in refining team line-ups, offering valuable insights for coaches and strategists to enhance player combinations and overall performance in competitive sports.

Previous studies have applied statistical and stochastic modelling to sport performance, using regression models to study performance indicator-outcome links in basketball games (Dong et al., 2021; Nakai et al., 2019; Sampaio, Lago-Peñas, et al., 2013). However, these approaches lack match evolution details. The proposed Multiple Criteria Decision Analysis (MCDA) approach enables comprehensive teams and player performance evaluation by considering multiple indicators and criteria (Haimes & Hall, 1974; Jiménez Martín et al., 2012; Trost, 2022). For example, Kizielewicz and Dobryakova (2020) tackled the challenges of accurately ranking NBA basketball players amidst incomplete data, demonstrating the potential of the Collection of Order Models to Establish Transparency (COMET) method – a multi-criteria decision-making approach based on fuzzy logic. This method enables reliable rankings with expert involvement (Kizielewicz & Dobryakova, 2020). These methods hold potential advantages in addressing multivariable and multi-objective challenges.

Meanwhile, Markov models are useful for modelling multidimensional game data, as they effectively describe the evolution of future successive possession by each of the two competing teams. Kvam and Sokol (2006) presented a combined logistic regression and Markov chain model for predicting the outcome of NCAA games (Kvam & Sokol, 2006). Cervone et al. (2016) proposed an Expected Possession Value (EPV) model based on the Markov model to quantify the expected points in basketball games (Cervone et al., 2016). Stavropoulos et al. (2021) proposed a second-order Markov model to evaluate the effectiveness of players' interactions and tactical structures on the weak side of basketball offence, specifically focusing on off-ball screens (Stavropoulos et al., 2021). In conclusion, these techniques have proven to be a valuable tool for analysing and predicting the outcomes of basketball games, providing a more detailed understanding of the evolution of the match and the effectiveness of team strategies, decision making, player interactions and game outcomes.

However, it is worth noting that these studies have mainly focused on men's basketball players, and it is uncertain whether these findings can be generalized to female data. In other words, it remains unclear whether team line-ups that well-perform in men's basketball will yield similar results in women's basketball. As a result, it is important for researchers to focus on how women's basketball players perform together in different line-ups, and to identify the key performance indicators that are the most effective in evaluating the efficiency of different player combinations in women's basketball.

As a result of the gap in knowledge identified above, the purpose of the present study was to investigate the synergies of women's basketball teams by analysing the effectiveness of their five-player line-up performance. This paper also aims to extend its investigation into the Chinese women's basketball team's line-up performance during international high-level tournaments, confronting diverse opponent capabilities. Additionally, it forecasts the stability and optimization potential of various team line-ups in offensive performance, employing a Markov model to compare game performances and evaluate offensive effectiveness over time.

2. Materials and methods

2.1. Sample

The study adopted an observational protocol by collecting data related to 23 randomly chosen games played by 13 women's basketball teams that competed in the 2016 Rio (4 matches) and 2020 Tokyo Olympics (4 matches), as well as the 2018 Spain (7 matches) and 2022 Australia FIBA Women's Basketball World Cup (8 matches), totalling 3,387

possessions. The line-up of each team on the court was manually recorded for each possession, along with variables such as the final ranking of each team under the tournament and the score of each possession between the two teams. All procedures were formally approved by the Local Institution of Research Review Board.

2.2. Procedure

Figure 1 illustrates the comprehensive framework adopted in this study. The first stage of our study was to extract each game possession based on play-by-play data instead of relying on video analysis. Ball possession in basketball refers to the period of play during which one team controls the ball until the opposing team gains possession (Sampaio & Janeira, 2003). A possession starts when one team gains control (or possession) of the basketball and ends when that team gives up control of the basketball. Teams can give up possession of the basketball in several ways, including (1) made field-goals or free-throws that lead to the other team taking the ball out of bounds, (2) defensive rebounds, and (3) turnovers. Notably, an offensive rebound does not start a new possession; an offensive rebound starts a new play (Kubatko et al., 2007). Therefore, the number of possessions per game was calculated using the occurrence times of specific game events.

Teams' line-up data were obtained by recording and digitizing the different five-player line-ups by watching videos of each game. In this research, players were classified into 7 positions, including Point Guard (PG), Shooting Guard (SG), Guard (G), Small Forward (SF), Power Forward (PF), Forward (F), and Centre (C). A G refers to a player who can handle the ball like a PG and score like a SG, while a F refers to a player who can play both SF and PF positions. Therefore, possible line-ups included all combinations of player positions, such as PG-SG-SF-PF-C, G-SF-F-PF-C, and so on.

2.3. Data reliability

To ensure the accuracy of the data, three researchers in basketball performance analysis independently observed possession and line-up data for three randomly selected full games. Each researcher has an experience of around 10 years in basketball training, and 3 years of practicing performance analysis in this sport. The observed gap spanned a two-week period, covering all three games and involving all three researchers. Cross-validation results showed that the Intra-Observer kappa coefficient was between [0.95, 1.00], indicating high agreement within the group of observers. The Inter-Observer kappa coefficient was between [0.93, 1.00], indicating high agreement between the observers.

2.4. Statistical analysis

2.4.1. Evaluating line-up efficiency

Offensive rating (OFR), defensive rating (DFR), and net rating (NR) are commonly used to describe per-possession efficiency. Offensive rating refers to the number of points scored per 100 possessions, defensive rating refers to the number of points allowed per 100 possessions, and net rating represents the average point differential per 100 possessions (Kubatko et al., 2007). By calculating these ratings of different team line-ups, we can compare the performance of each line-up.

$$OFR_f = \frac{PTS_f}{POSS_f} \times 100 \tag{1}$$

$$DFR_f = \frac{PTS_o}{POSS_o} \times 100 \tag{2}$$

$$NR_f = OFR_f - DFR_f \tag{3}$$

Above *PTS* means the total points scored by the team and subscript *f* denotes the line-up of a team, and subscript *o* denotes the opponent team. Our research objectives centre on exploring offensive performance dynamics in-depth. OFR aligns with this focus, allowing us to uncover nuanced insights into offensive strategies, player interactions, and the impact of player substitutions, while maintaining a clear and targeted approach to our analysis. Therefore, offensive rating was categorized into discrete variables of three intervals (Kolias et al., 2022). As a result, these three intervals reflect three different offensive performance levels of teams' line-ups: low efficiency, medium efficiency, and high efficiency (Charamis et al., 2023; Clay & Clay, 2014).

$$S = \begin{bmatrix} 1 = (0 - 80), \\ 2 = (80 - 120), \\ 3 = (120 - max) \end{bmatrix} \tag{4}$$

2.4.2. Clustering for team quality classification and identifying frequently used team line-ups

Firstly, we used the k-means clustering algorithm to classify the quality of each team. During modelling, we used the k-means clustering algorithm with k = 3. All team rankings were categorized into three groups based on their final standings in the competitions: 1st to 5th place (high quality), 6th to 10th place (medium quality), and over 10th place (low quality). This modelling process was conducted using IBM SPSS Statistics 28.

Also, the DBSCAN clustering algorithm to classify the team line-ups in order to filter out the most common used team line-ups. The DBSCAN algorithm typically requires two parameters: *eps* and *min samples*. To identify the most common-used line-ups, we performed the DBSCAN algorithm using two features: the possessions (POSS) and points for each line-up (Points), with the hyperparameters *eps* = 0.3, and *min samples* = 3.

2.4.3. Definitions of the discrete-time Markov model

In what follows we provide a brief overview of discrete-time Markov models. A Markov chain model is a stochastic process that consists of a *state space(S)* and t*he transition probabilities* p_{ij}. In our case, we defined a state space that reflects the three different levels of the line-up's OFR, as shown in (4) above.

In addition, p_{ij} equals the probability of transitioning from the i^{th}offensive rating level to the

j^{th} for the successive reappearances of the same line-up. For instance, p_{13} is the probability that a line-up starting from the first offensive rating level (0–80) moved to the third rating level (120 – max) at its next reappearance. For those probabilities, the Markovian property holds as the information contained in

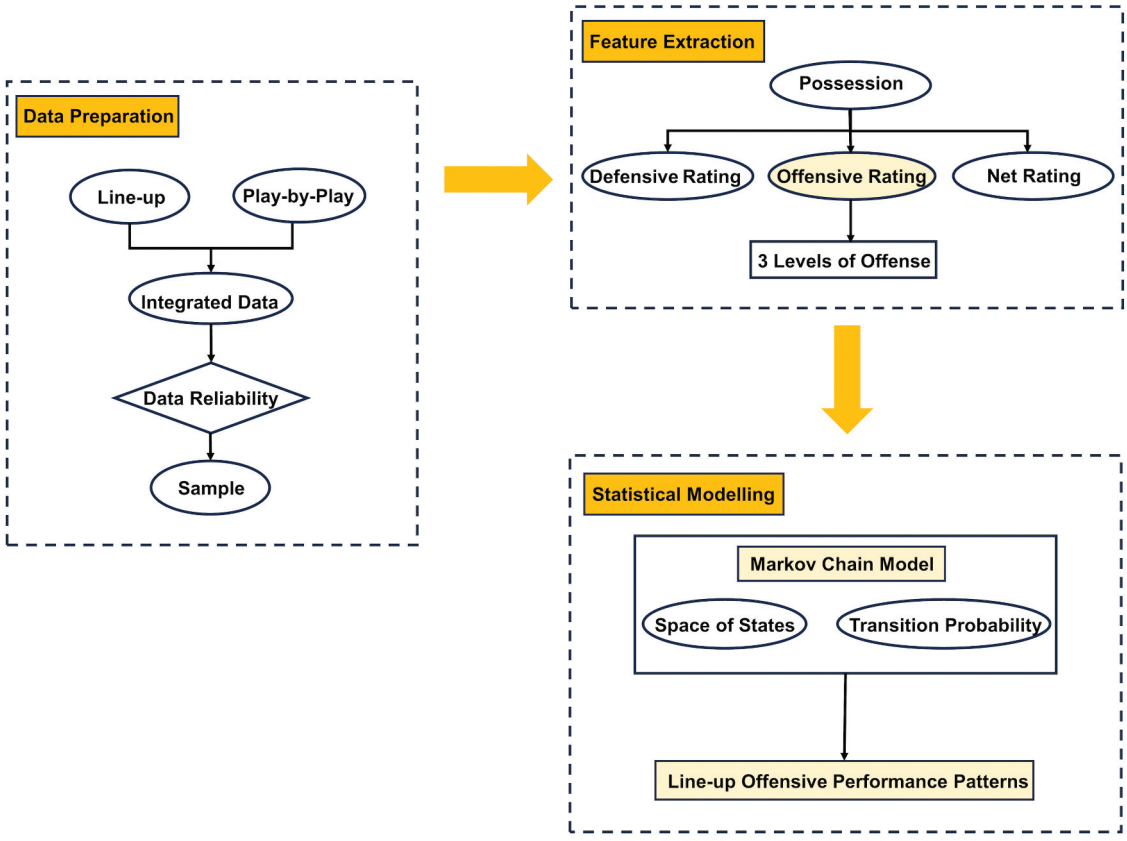


Figure 1. A graphical representation of the proposed framework.

the current state of the process is all that is needed to determine the probability distribution of the next states. This property, called Markovian, could be written as:

$$P(X_{n+1}=j|X_n=i,X_{n-1}=i_{n-1},\ldots,X_0=i_0)=P(X_{n+1}=j|X_n=i)=p_{ij}(n) \quad (5)$$

Where X_n denotes the offensive rating level of the line-up during its nth appearance in court.

The probabilities p_{ij}, as described above, form the stochastic matrix P (transition probability matrix of the Markov chain), which includes the information of the transitions between offensive rating levels.

Furthermore, given that the transition probability matrix is irreducible and aperiodic, the

asymptotic distribution is defined as the initial probability vector $\pi=[\pi_0,\pi_1,\pi_2,\ldots,\pi_n]$,where

$$\pi_j=\lim_{n\to\infty}\left(P_{ij}\right)^n, j\in S \quad (6)$$

The value π_j is the long-run probability that the specific line-up is evaluated in offensive rating level *j*. The asymptotic distribution satisfies the following system of equations:

$$\sum_{j\in S}\pi_j=1 \text{ and } \pi P=\pi \quad (7)$$

Taken together, a Markov chain is fully determined by an initial probability vector π_0 and the respective transition matrix *P*. Following a previous recommendation by (Kolias et al., 2022), the current study combined 2 basic parameters, *line-up optimization expected number of player rotations* and *line-up stabilization expected number of player rotations.*

Line-up optimization expected number of player rotations (***optimization***$_{ij}$, ***o***$_{ij}$)denotes the average expected number of player rotations required for a team's line-up to transition from the current state of offensive (i) level to the higher offensive level (j).

$$optimization_{ij}=\frac{1}{p_{ij}} \quad (8)$$

Line-up stabilization expected number of player rotations (***stabilization***$_i$, ***s***$_i$) denotes the average expected number of player rotations for the team's line-up to consistently remain at the current offensive level (i).

$$stabilization_i=\frac{1}{1-p_{ij}} \quad (9)$$

In both equations above, p_{ij} denotes the transition probability of the line-up in its current offensive state. From the standpoint of team performance, a lower value for optimization$_{ij}$ means that slight rotation in the line-up can lead to an improved offensive performance. On the other hand, a higher value for stabilization$_i$ indicates that the line-up possesses a robust ability to sustain its current offensive level, and is not easy to be transfer to other levels by player rotation adjustment.

3. Results

A total of 155 potential line-ups were identified across all 23 games by 13 competing teams. Among them, 33 line-ups were

used by the Chinese national women's basketball team, while 90 were used by other teams, and 32 were co-used by the Chinese national team and other teams. Achieving a silhouette coefficient of 0.65 validates the effectiveness of the clustering process, as depicted in the preceding Figure 2. Based on the clustering results, 12 representative line-ups that the Chinese team used over 100 possessions (POSS) were selected for further research. Additionally, 10 representative line-ups used by other competing teams were determined for further research.

Table 1 displays the mean Offensive Rating (OFR), Defensive Rating (DFR), and Net Rating (NR) for each Chinese team line-up, along with the number of possessions and the game year in which they were used. From a general perspective, it is evident that all of the 12 most frequently used line-ups by the Chinese team include a Centre position. From the perspective of possessions used, the most frequently utilized line-up was G-SG-SF-C-C, which was employed for a total of 355 possessions and included two Centre players. From the perspective of on-court performance, the PG-G-SF-PF-C and G-SG-SF-F-C line-ups had notably high offensive efficiency, with mean OFR values of 133.93 and 127.45, respectively. A lower value of DFR indicates better defence, and the line-ups with the highest-performing defence were PG-G-SF-PF-C, followed by G-G-F-PF-C and PG-G-SF-C-C, with mean DFR values of 76.6, 77.27, and 84.62, respectively. Interestingly,

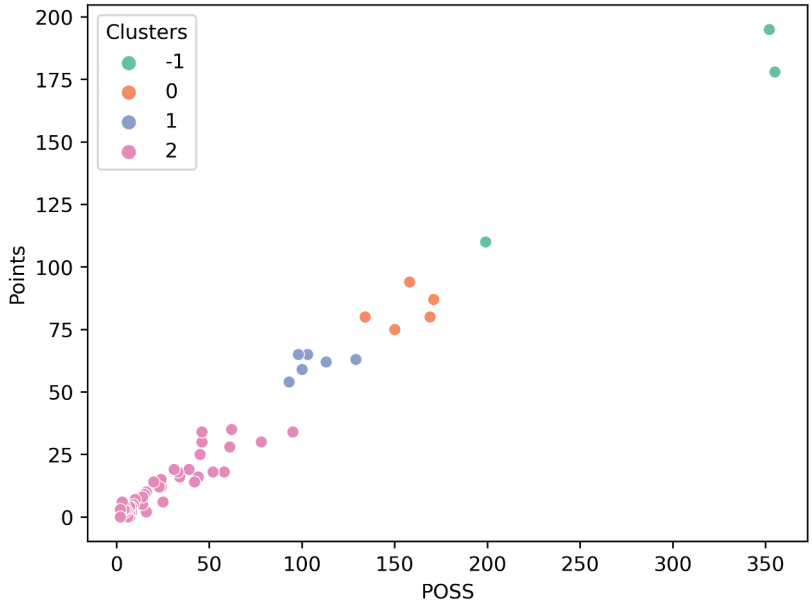


Figure 2. The clustering results of the observed line-ups.

Table 1. Descriptive statistics of mean OFR, DFR, and NR for each observed Chinese team line-up.

Line-ups	Year	Team	POSS	POSS (*normalized*)	Mean OFR	Mean DFR	Mean NR
G-SG-SF-C-C	2018 2020	CHN	355	10.48%	109.2	103.65	5.55
PG-G-SG-PF-C	2018 2020 2022	CHN	351	10.36%	115.38	89.07	26.31
PG-G-F-PF-C	2018 2020 2022	CHN	199	5.88%	110	108.08	1.92
G-SF-SF-C-C	2016	CHN	170	5.02%	101.1	91.76	9.4
G-SG-SF-PF-C	2018 2020	CHN	169	4.99%	108.11	89.47	18.64
PG-G-SF-C-C	2018 2020	CHN	158	4.67%	117.5	84.62	32.88
PG-G-SF-PF-C	2016 2018 2020	CHN	150	4.43%	133.93	76.6	57.33
G-G-F-PF-C	2016 2022	CHN	134	3.96%	117.64	77.27	40.37
PG-G-F-C-C	2018 2022	CHN	129	3.81%	100	116.67	−16.67
PG-G-SG-F-C	2020 2022	CHN	113	3.34%	108.77	96.43	12.34
G-SG-SF-F-C	2020	CHN	103	3.04%	127.45	101.92	25.53
G-SF-F-PF-C	2016	CHN	100	2.98%	111.32	110.64	0.68

OFR: Offensive Rating; DFR: Defensive Rating; NR: Net Rating.

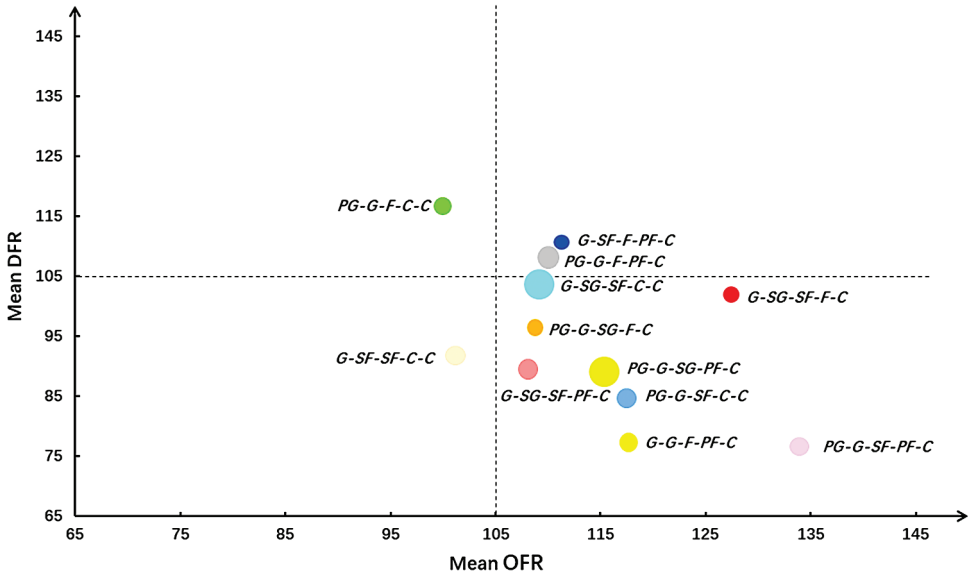


Figure 3. Bubble plot of each observed Chinese line-up. Note: each bubble denotes a specific line-up, whose size denotes the number of possessions for each line-up. A larger bubble indicates that the line-up had more possessions.

PG-G-SF-PF-C had the highest mean NR, as it had both high OFR (133.93) and DFR (76.6).

Figure 3 presents the distribution of different line-ups considering their mean OFR and DFR. The area in the lower right corner of the diagram represents a line-up with relatively good offensive and defensive performance, where most of the line-ups are located (8 out of 12 line-ups). In contrast, the line-up PG-G-F-C-C in the upper left corner showed the relatively lowest offensive and defensive performance (NR = −16.67).

Table 2 presents the mean OFR, DFR, and NR for 12 countries, which are China's opposing teams. Despite its modest offensive rating (Mean OFR:89.11) and defensive ratings (Mean DFR:91.93), the PG-SG-SF-PF-C line-up remains popular among opponent teams, notably used by European countries like France and Belgium. In addition, Japan's usage of PG-SF-SF-SF-PF boasted the highest offensive rating (Mean OFR: 120). Lastly, Team USA and Australia both favour three-guard line-ups like G-G-G-F-C (Mean OFR: 89.77) and G-G-G-F-PF (Mean OFR: 98.19).

Table 3 displays varying mean OFR values for different Team China line-ups against opponents of diverse qualities. Comparing these line-ups, PG-G-SF-C-C and G-SF-F-PF-C have higher offensive efficiency when facing stronger opponents, with Mean OFRs of 116.55 and 128.03, respectively. G-SG-SF-F-C and PG-G-F-PF-C perform well against moderate opponents, with Mean OFRs of 149.38 and 128.75 respectively. G-G-F-PF-C, G-SG-SF-F-C, PG-G-SG-F-C, PG-G-F-PF-C, and PG-G-SG-PF-C are efficient against weaker opponents, with Mean OFRs of 162.5, 140.00, 135.47, 134.81 and 131.94, respectively. Notably, PG-G-SF-C-C sustains stable offensive performance across different opponent strengths (Mean OFR: 116.55, 116.55, 112.70). Sample size constraints precluded identifying performance against moderately and weaker-powered teams for a few line-ups.

Table 2. Mean OFR, DFR, and NR for the line-ups of opponents.

Line-ups	Year	Team	POSS	POSS (*normalized*)	Mean OFR	Mean DFR	Mean NR
PG-SG-SF-PF-C	2018 2020	FRA;BEL; SRB;LAT	264	7.79%	89.11	91.93	−2.82
PG-SG-F-PF-C	2016 2018 2022	FRA;PUR; USA;SRB; SEN	134	3.96%	100.41	111.61	−11.2
PG-F-PF-C-C	2016 2018	SEN	133	3.93%	88.74	101.43	−12.69
G-G-F-PF-PF	2022	USA	123	3.63%	99.27	74.68	24.59
PG-PG-SF-C-C	2022	BEL	118	3.48%	85.71	98.39	−12.68
PG-SF-SF-SF-PF	2018	JPN	116	3.42%	120.00	119.64	0.36
G-G-G-F-C	2018 2020 2022	USA;AUS KOR; CAN;	107	3.16%	89.77	123.15	−33.38
PG-G-SG-PF-C	2020 2022	PUR;BIH; SRB	106	3.13%	97.82	126.35	−28.53
G-G-G-F-PF	2020 2022	USA;AUS	89	2.63%	98.19	113.40	−15.21
PG-SF-F-PF-PF	2020 2022	PUR;BEL	78	2.30%	116.18	88.57	27.61

OFR: Offensive Rating; DFR: Defensive Rating; NR: Net Rating.

Table 3. Mean OFR for each Chinese line-up against the different quality of opponent.

	Opponent quality		
Line-ups	high	medium	low
G-SG-SF-C-C	74.88	109.55	61.43
PG-G-SG-PF-C	99.88	114.99	131.94
PG-G-F-PF-C	78.38	128.75	134.81
G-SF-SF-C-C	97.91	–	125.00
G-SG-SF-PF-C	59.30	110.06	115.00
PG-G-SF-C-C	116.55	116.55	112.70
PG-G-SF-PF-C	83.33	103.51	106.88
G-G-F-PF-C	113.81	116.43	162.50
PG-G-F-C-C	100.88	100.00	66.67
PG-G-SG-F-C	81.86	–	135.47
G-SG-SF-F-C	100.00	149.38	140.00
G-SF-F-PF-C	128.03	–	–

Remarkably, to attain the highest offensive level (120-max), this particular line-up requires only an average of 1.67 player rotations. These findings suggest that in a given game scenario, when the offensive rating (OFR) of the G-SG-SF-C-C line-up is below 120, coaches may need a minimum of 5 player rotations (averaging over 4.55) to enhance the likelihood of reaching a higher offensive rating. Conversely, the G-G-F-PF-C line-up requires only 2 player rotations (averaging over 1.67) to achieve a higher offensive rating.

4. Discussion

Through a novel approach, the current article delved into the profound insights underlying the game performance of different female basketball line-ups during gameplay. To begin with, we accomplished the collection and construction of line-up data for a total of 23 international games of 14 women's basketball teams, including Chinese basketball team and 13 other opposing teams. These selections were made to investigate the utilization disparities between China's women's basketball team and those of other nations in international competitions, aiming to optimize team performance through strategic line-up choices. Then we completed the data reliability verification, the final Kappa coefficient is above 0.90, this result shows that the data constructed by the research has a strong reliability. Subsequently, we conducted computations for offensive rating (OFR), defensive rating (DFR), and net ratings (NR) pertaining to each team line-up, thereby undertaking a comprehensive assessment of their in-game performance. Additionally, we categorized the team's offensive rating (OFR) into three levels and evaluated the expected number of player rotations for a team's line-up to maintain its current offensive performance and optimize its current offensive level. Furthermore, we explored the offensive performance of the various Chinese line-ups against opponents of varying quality. Based on the work on feature index extraction and statistical model construction, this paper obtained the convergence of team line-ups between different offence levels and the evolution of line-ups in terms of offensive efficiency in basketball games from both retrospective and predictive perspectives. These findings revealed the performance of different line-ups in different game scenarios and help coaches make more informed decisions in female basketball.

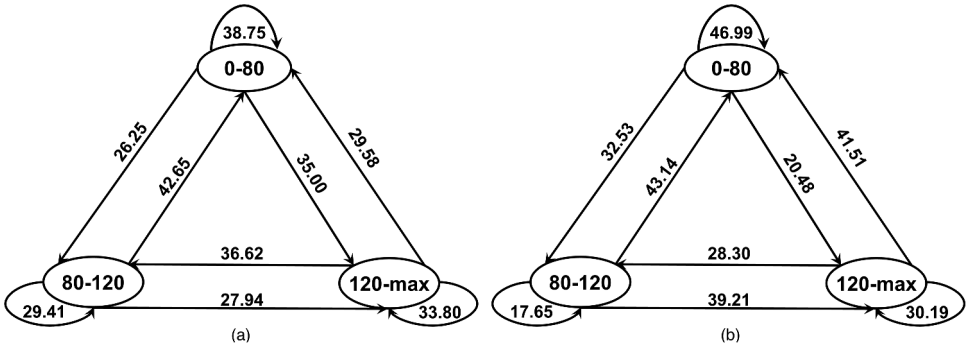


Figure 4. Markov chain transition diagram between OFR levels (a: China; b: Opposing teams).

The transition probabilities between three offensive rating levels for team China and its opponents are presented in Figure 4. Overall, the offensive performance of the Chinese line-up was relatively consistent, maintaining a transition probability of around 30% between all different offensive levels Figure 4(a). While the offensive performance of the opposing team line-ups tends to fluctuate, with a range of transition probabilities between 17% and 47% Figure 4(b).

Figure 5 presents the probabilities of transitioning between OFR levels for each of the included line-ups, along with the ***stabilization***$_i$ (s_i) for each level. On the one hand, the 4th line-up, G-SF-SF-C-C, starts from the lowest OFR level (0–80), has zero probability of transitioning to the highest OFR level (120-max), and an 80% probability of remaining in the same OFR level (0–80). On the other hand, the 5th line-up, G-SG-SF-PF-C, which also starts from a low OFR category, has a 60% probability of transitioning to the high OFR category at its next appearance in the game.

Table 4 provides the asymptotic vector, represented by the symbol π, and the asymptotic mean ***optimization***$_{ij}$ times (o_{ij}) of line-ups for each offensive rating (OFR) level, indicates in parentheses. These asymptotic probabilities indicate the long-term likelihood of a line-up residing in a specific OFR level during the game. For instance, the asymptotic probability that the offensive rating distribution of the line-up G-SG-SF-C-C is at the low (0–80), medium (80–120), and high (120-max) offensive levels is 0.29, 0.49, and 0.22, respectively. Conversely, for line-up G-G-F-PF-C, the asymptotic probabilities of the distribution of offensive rating at the low (0–80), medium (80–120), and high (120-max) levels of offence are 0.20, 0.20, and 0.60, respectively.

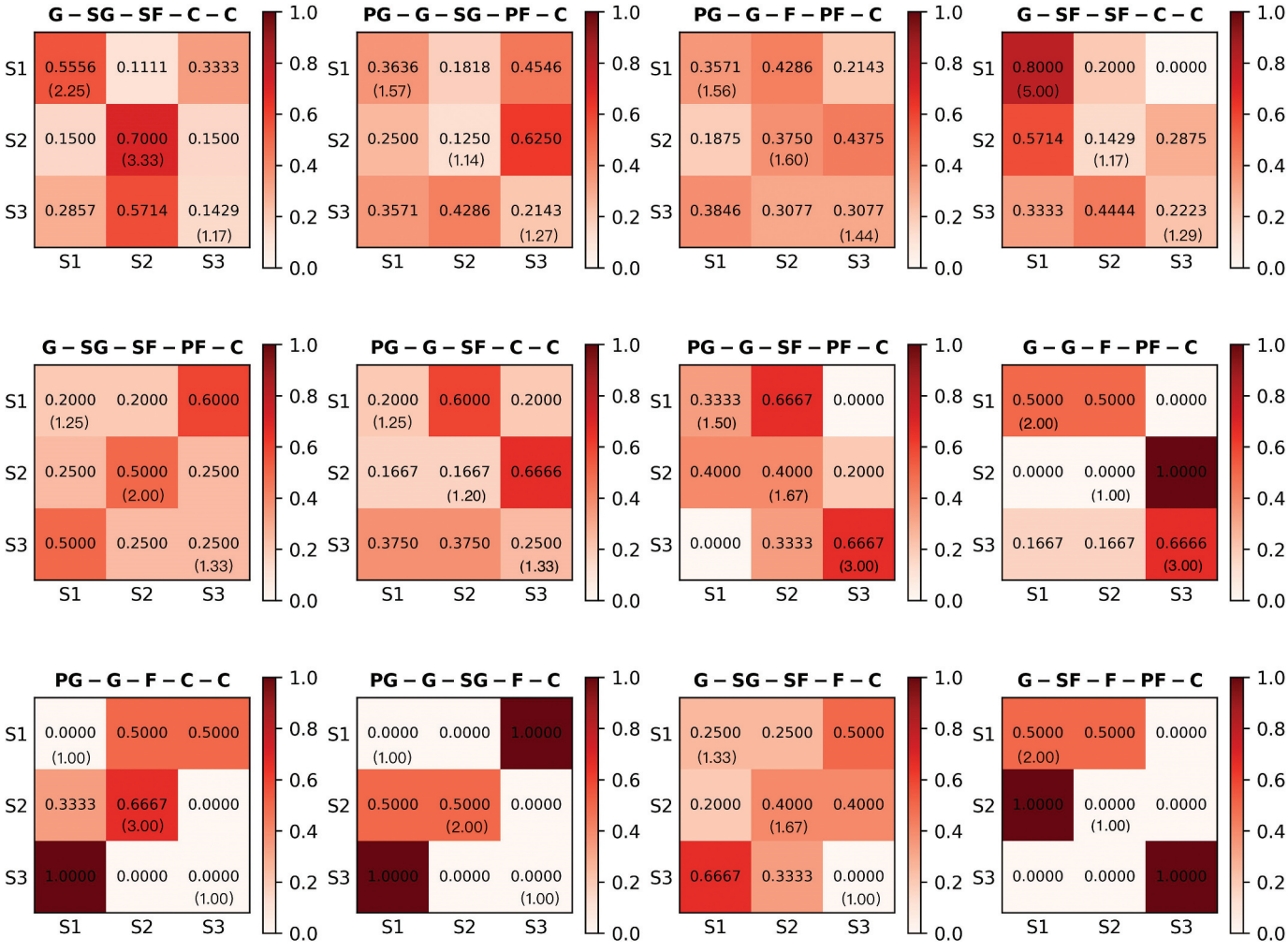


Figure 5. Transition probabilities between offensive levels for different team line-ups. Note: each heatmap represents the transition probabilities of a team's line-ups between three offensive levels. Darker colours indicate higher probabilities of transition between offensive levels for specific line-ups.

Table 4. Asymptotic probability distribution and mean times for each Chinese line-up.

	Offensive rating levels		
Line-ups	0 ~ 80	80 ~ 120	120~max
Total	0.38(2.63)	0.32(3.13)	0.30(3.33)
G-SG-SF-C-C	0.29(3.45)	0.49(2.04)	0.22(4.55)
PG-G-SG-PF-C	0.33(3.03)	0.27(3.70)	0.40(2.50)
PG-G-F-PF-C	0.30(3.33)	0.37(2.70)	0.33(3.03)
G-SF-SF-C-C	0.72(1.39)	0.21(4.76)	0.07(14.28)
G-SG-SF-PF-C	0.32(3.13)	0.32(3.13)	0.36(2.78)
PG-G-SF-C-C	0.26(3.85)	0.36(2.78)	0.38(2.63)
PG-G-SF-PF-C	0.27(3.70)	0.46(2.17)	0.27(3.70)
G-G-F-PF-C	0.20(5.00)	0.20(5.00)	0.60(1.67)
PG-G-F-C-C	0.33(3.03)	0.50(2.00)	0.17(5.88)
PG-G-SG-F-C	0.47(2.13)	0.16(6.25)	0.37(2.70)
G-SG-SF-F-C	0.37(2.70)	0.32(3.13)	0.31(3.22)
G-SF-F-PF-C	0.45(2.22)	0.23(4.35)	0.32(3.13)

Our analysis revealed that the G-SG-SF-C-C is the most commonly frequently employed line-up for the Chinese women's basketball team is, notable for its inclusion of two centre players. Other commonly observed line-ups include PG-G-SG-PF-C, PG-G-F-PF-C, and G-SF-SF-C-C. When confronted by opponents of varying strengths, these frequently employed team line-ups commonly manifest a recurring pattern: confronting formidable adversaries results in decreased offensive efficiency, encountering opponents of intermediate strength yields relatively stable offensive efficiency, and facing weaker opponents leads to a slightly above-average offensive efficiency. However, surprisingly, the extensively used G-SG-SF-C-C line-up demonstrates notably diminished offensive efficiency against weaker opponents, characterized by a mean OFR as low as 61.43. This phenomenon could be attributed to: i) the line-up's extensive temporal scope, often employed in events dating back to a time when global women's basketball was still in its developmental stages, resulting in the prevalence of offensive inefficiency among most teams; ii) moreover, the dual centre composition of China's team line-up itself displays an irregular offensive efficiency trend – prospering against stronger adversaries but struggling against weaker ones – prompting strategic considerations for the judicious deployment of this line-up by the coaches (Bazanov & Rannama, 2015; Campaz, 2023; Mohamed et al., 2022). In contrast, for other competing teams, the most commonly utilized line-ups were PG-SG-SF-PF-C, PG-SG-F-PF-C, and PG-F-PF-C-C. However, despite the prevalence of two-tower line-ups, featuring two centres, our findings suggest that their frequency does not necessarily correspond to high efficiency in either offence or defence (Zarić et al., 2020).

Previous research has shown that important timeouts or player rotations during different game quarters can significantly influence a team's performance indicators, ultimately shaping the outcome of the match (Clay & Clay, 2014; M. A. Gómez et al., 2011; M.-Á. Gómez et al., 2017; Siegel & Daniel, 2019). However, given the 10-minute duration of each quarter in a basketball game, coaches still face a tough task in making crucial rotation decisions at specific points or contexts in the game. Our current findings might provide a predictive dimension to the coaching decision-making process, specifically addressing the nuanced dynamics of player rotations within distinct line-up configurations. For certain team line-ups, such as those involving guards (G), small forwards (SF), and centres (C) with combinations like G-SF-SF-C-C, PG-G-F-C-C, and G-SG-SF-C-C, a dynamic strategy of continuous player rotations becomes imperative. This strategic approach aims to disrupt the opposing team's defensive schemes, thereby optimizing the overall offensive efficiency of the team. Conversely, for line-ups like G-G-F-PF-C, PG-G-SG-PF-C, and PG-G-SG-F-C, coaches are advised to exercise restraint in player rotations. In these instances, a patient approach is recommended, allowing the team sufficient time to settle in the game's rhythm and consequently improve its overall performance. This analytical framework aids coaches in strategically optimizing offensive efficiency, thereby influencing the trajectory of the game in a more deliberate and informed manner.

Generally, basketball players can be classified into two main groups: inside or post players. Inside players include positions of PF and C, while perimeter players include the PG, SG, and SF positions (Trninić & Dizdar, 2000). In our study, the PG-G-SF-PF-C and G-SG-SF-F-C line-ups are the most commonly used line-ups, exhibiting the highest offensive rating in the Chinese women's national basketball team. Particularly, the PG-G-SF-PF-C combination dominated in both offensive and defensive performance, achieving the highest net rating of 57.33. We can infer that line-ups with a well-balanced combination of inside and perimeter players, showcasing better performance in possessions, shooting, and screen ability, result in a comprehensive and effective strategy. Additionally, the PG-G-SF-C-C line-up has been found to be the most efficient offensive and defensive line-up, with more than 150 possessions to its credit. Therefore, it can be inferred that the Chinese team would benefit from using a two-centre line-up in international competitions (Li et al., 2017). On offence, a centre player needs to be able to create their own shot in cramped spaces, secure rebounds, and be proficient with playing with their back to the basket. Centres must also collaborate effectively with teammates by setting up screens. On defence, the centre is typically the best shot blocker and a solid rebounder, crucial for preventing the opponent's second-chance baskets (Delextrat et al., 2015; Dezman et al., 2001; Fewell et al., 2012; Kalman & Bosch, 2020; Köklü et al., 2011; Sampaio et al., 2006; Starkes et al., 1994; Trninić & Dizdar, 2000). Different combinations of the five-player positions bring various game performances to the team (Reina Román et al., 2019). Understanding each position's responsibilities can help coaches formulate effective strategies and make informed decisions regarding player selection and game plans (Sandholtz et al., 2020).

Regarding China's opponents, the 2020 Japanese team used a PG-SF-SF-SF-PF line-up, which had very high offensive efficiency and helped them win a silver medal in the Tokyo Olympics women's basketball tournament. Although Japanese players have a height disadvantage in the C position compared to other teams, their line-up benefits from having three players in the SF position, with excellent mid-long range shooting ability and fast break ability (Huang, 2022).

The most notable defensive line-up was the G-G-F-PF-PF used by Team USA in 2022. It is surprising to note that this line-up did not include a C player, which challenges the traditional notion that big men are essential for protecting the basket during games (Sampaio, Ibáñez, et al., 2008). Basketball is now considered a more position-less sport, and the forward position has become a multi-functional player position that not only has a defensive ability comparable to that of a C to defend the basket but also has more flexible footwork and movement speed (Kalman & Bosch, 2020). Previous studies have indicated that F and G have a positive impact on defence, with F contributing more defensive rebounds and committing fewer fouls resulting in free throws, while G has a higher potential for defensive rebounds, more steals, and fewer fouls resulting in free-throws on defence (Choi et al., 2015). Hence, we can conclude that a combination of G and F tends to have a significant defensive impact.

The present study faces methodological limitations in assessing team performance and exploring the suitability of finite Markov chains. In the player position division, a notable limitation arises from the absence of player size information, potentially impacting their roles on the court. Our categorization was exclusively based on official player profiles, neglecting individual characteristics within the same position in the line-up. To improve our approach, future work should consider a more nuanced and flexible classification of on-court positions, taking into account the technical characteristics of each player rather than rigidly adhering to predefined roles. Also, it is essential to acknowledge the inherent variability in line-ups across different tournaments. While our analysis primarily relied on sampling line-up observations from multi-year competition data to discern patterns and trends, it is crucial to note that team compositions may undergo changes between tournaments. Player personnel may vary due to factors such as injuries or team strategy adjustments. In subsequent iterations, there are plans to comprehensively explore team dynamics, with a specific focus on team cohesion, communication, and synergy. The aim is to strategically cultivate these aspects, minimizing disruptions during player rotations and sustaining a consistent level of performance (Eys et al., 2019). Ultimately, this paper solely delved into modelling and probing the stability and optimization of line-ups through an offensive lens, without extending its scope to encompass defensive aspects and the overall combined performance of the team in both offensive and defensive. Subsequent research endeavours could augment these dimensions for further exploration and analysis, providing coaches with a more comprehensive suite of decision-making and analytical tools.

In summary, while the present study provides valuable insights into the effectiveness of different female basketball team line-ups, further research is needed to address

the methodological limitations and to provide a more comprehensive understanding of the relationship between team line-ups, offensive and defensive strategies, and game outcomes in basketball and other team sports.

5. Conclusion

The results of the current study demonstrate the usefulness of the Markov chain model combined with key performance indicators in evaluating the match performance of different team line-ups. Our findings suggested that: (i) Teams from different countries tend to have their own preferences in terms of line-up usage tendencies, player position combinations, and offensive and defensive efficiency; (ii) Different line-ups exhibit considerable variability in terms of their stability and directional tendencies in the offence. These findings provide insightful ideas for coaches, highlighting the importance of considering various game scenarios, such as the quality of the opposing team and their offensive or defensive characteristics when selecting a line-up. Furthermore, it is possible that performance analysts in team sports use this method to model defensive performance for specific team line-ups under multiple game contexts.

Disclosure statement

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